Thin Film Growth Simulator

An interactive look at how materials take shape

Code

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Interactive application

Open the Streamlit app

Summary

Thin films are the building blocks of many electronic devices. Their quality depends on how material settles onto a surface—and whether it reaches the small features underneath.



The simulator brings the process, the growing film and the final inspection together in one view.

1 The project

This interactive simulator makes an invisible process visible. Watch a film grow, explore how different conditions change its shape, and see why a layer that looks solid can still hide an empty pocket below the surface.

2 Explore it yourself

- **Choose a recipe.** Adjust the conditions and the shape of the surface.
- **Watch the growth.** Follow the process and replay how the film forms.
- **Compare the outcome.** Inspect the finished layer and see how a different recipe changes the result.

3 Why it matters

The project turns an abstract materials-science problem into a visual, hands-on experience. It connects scientific ideas with the practical challenge of making reliable films, inviting users to learn through experimentation.

Try the live simulator